

Question:

A 35-year-old African American man presents with fever, abdominal pain, and severe weakness since yesterday. On physical examination, the patient is jaundiced and shows a generalized pallor. Past medical history is significant for recently receiving anti-malaria prophylaxis before visiting Nigeria. Laboratory tests show normal glucose-6-phosphate dehydrogenase (G6PD) levels. Peripheral smear shows the presence of bite cells and Heinz bodies. Which of the following is the most likely diagnosis in this patient?

Answer Choices:

- A. Autoimmune hemolytic anemia
- B. Microangiopathic hemolytic anemia
- C. Paroxysmal nocturnal hemoglobinuria (PNH)
- D. Glucose-6-phosphate-dehydrogenase (G6PD) deficiency

Reasoning + Answer:

<think> Reasoning process here: The patient presents with symptoms of hemolysis, such as jaundice and pallor, and has a history of anti-malaria prophylaxis, which suggests exposure to a malaria parasite. The presence of bite cells and Heinz bodies on the peripheral smear is indicative of oxidative stress, which is consistent with a deficiency in an enzyme that protects red blood cells from oxidative damage. The normal G6PD levels in this patient make G6PD deficiency less likely. The patient's symptoms and laboratory findings are not consistent with autoimmune hemolytic anemia or PNH, which typically present with other distinct clinical features. </think><answer> B. Microangiopathic hemolytic anemia </answer>

